
Clinical Trial Funding Application v2.0

NIH-adaptable concept and submission template

Preparation note. This package is a drafting and funder-outreach template, not an official substitute for an agency portal, SF424 forms, ASSIST, Grants.gov Workspace, or a funder's required application system. Verify the live solicitation before submitting a new application. [10.5281/zenodo.21317266](https://doi.org/10.5281/zenodo.21317266)

Target funding organization

Up to 160 chars

National Institutes of Health

Program / award title

Up to 180 chars

NIH Director's Pioneer Award

Custom opportunity, solicitation, call, or program number

Up to 100 chars

RFA-RM-27-001

Proposed project title

Up to 220 chars

Daraxonrasib Phase 1 LLM-Advised Robotic Whipple Trial in KRAS-Mutated PDAC

Applicant / PD-PI

Kevin Kawchak, M.S. (Chemistry)

Applicant organization

ChemicalQDevice

Applicant email

kevink@chemicalqdevice.com

Telephone

619-371-0010

Submission / transmittal date

July 12, 2026

Requested project period

5 years

Transmittal message

Recipient name, title, and organization

Up to 500 chars

Patricia Labosky, Ph.D., Program Leader, NIH Common Fund, Office of Strategic Coordination, Office of the Director, National Institutes of Health

Professional transmittal / cover message

Up to 2400 chars

Dear Dr. Labosky:

ChemicalQDevice respectfully submits “Daraxonrasib Phase 1 LLM-Advised Robotic Whipple Trial in KRAS-Mutated PDAC” to the NIH Director’s Pioneer Award. The application proposes a first-in-human, combined drug-device investigation of perioperative daraxonrasib and an eight-arm robotic pancreaticoduodenectomy, with an on-premises, repository-pinned LLM acting only as a second-opinion advisory system [[Kawchak, 2026a](#); [Kawchak, 2026c](#)].

Three bounded generation roles are tested: synchronized drafting of protocol, consent, regulatory, and operating artifacts; task-level surgical advisories derived from verified sensor context; and perioperative daraxonrasib pause-and-restart recommendations. Every output is source-, model-, commit-, and seed-pinned, screened before generation by verification, validation, and uncertainty-quantification gates, written to an audit trail, and accepted only after qualified human review. The LLM cannot authorize treatment, enter the vendor kinematic stack, or command an actuator.

This is a new research direction rather than a routine scale-up: targeted therapy, robotic Whipple execution, human authority, Combined IND/IDE obligations, LLM peer review, and computational assurance are prospectively co-designed as one falsifiable clinical-trial system. The Phase 1 Protocol provides the operative, dosing, endpoint, and oversight context used throughout this application [[NIH, 2026a](#); [Kawchak, 2026b](#); [Kawchak, 2026f](#); [Kawchak, 2026g](#)].

Sincerely,
Kevin Kawchak
CEO, ChemicalQDevice

1 Opportunity and Applicant Organization

Preparation note. Use this section as the tailoring control sheet. The defaults reproduce key RFA-RM-27-001 parameters but every field remains editable for another U.S. federal agency, foundation, nonprofit, disease-focused organization, health system, or corporate sponsor.

Opportunity tailoring

Funder	National Institutes of Health
Program / award	NIH Director's Pioneer Award
Opportunity / solicitation number	RFA-RM-27-001
Mechanism / instrument	DP1 - Clinical Trial Optional
Clinical-trial status	Clinical Trial Optional
Due date / time zone	September 9, 2026, 5:00 PM local time
Submission route	NIH ASSIST
Research strategy limit	5 pages
Facilities limit	1 page
Budget ceiling	\$700,000 direct costs/year; \$3,500,000 total

Applicant organization identifiers

Legal organization name	Department / division
ChemicalQDevice	N/A
Unique Entity Identifier (UEI)	CAGE / NCAGE
Pending via SAM	Pending CAGE via SAM
Organization type	Tax / nonprofit status
LLC	For-Profit
Street address	City, state, ZIP
2869 Historic Decatur Rd. Suite 100	San Diego, CA 92106

Organization mission and fit with the target funder

Up to 1400 chars

ChemicalQDevice develops auditable methods for oncology trial design, regulatory preparation, simulation, and governed Physical AI. The proposed work fits the Pioneer mechanism because it tests a fundamental new premise: in patient-facing generation systems, verification must precede generation, and human clinical authority must remain outside the model. The Phase 1 program integrates targeted therapy, an eight-arm robotic Whipple, on-premises LLM advisories, Combined IND/IDE governance, and reproducible provenance as one high-risk, potentially generalizable research direction rather than a conventional device or drug scale-up [NIH, 2026a; Kawchak, 2026c; Kawchak, 2026b; Kawchak, 2026f].

Eligibility basis and registration status

Up to 1200 chars

ChemicalQDevice is presented as a domestic U.S. for-profit small-business applicant, an eligible organization type under RFA-RM-27-001 [NIH, 2026a]. Before submission, the live SAM registration, UEI, CAGE, Grants.gov registration, organizational eRA Commons account, PI eRA Commons affiliation, and PI ORCID must be active and mutually consistent. If the PI also serves as Signing Official, separate eRA Commons accounts are required for those roles. No foreign organization or foreign subaward is proposed; any foreign component or unfunded collaboration would require the permitted justification and disclosure.

Authorized organizational representative and grants-administration contacts

Up to 1100 chars

Kevin Kawchak, M.S., Chief Executive Officer, ChemicalQDevice
Authorized Organizational Representative / Signing Official
2869 Historic Decatur Rd., Suite 100, San Diego, CA 92106
phone: 1-619-371-0010
email: kevink@chemicalqdevice.com
eRA Commons IDs: pending verification
The PI and Signing Official roles must use distinct accounts if held by the same individual

2 Project Summary, Public Narrative, and Nontechnical Overview

Preparation note. For the NIH baseline, keep the Project Summary/Abstract to approximately 30 lines and the Project Narrative to three sentences. Other funders may impose word or character limits instead. The character caps below are conservative drafting controls and can be changed in the section source.

Project title

Up to 220 chars

Daraxonrasib Phase 1 LLM-Advised Robotic Whipple Trial in KRAS-Mutated PDAC

Primary disease, condition, population, or scientific area

Up to 180 chars

KRAS-Mutated Pancreatic Ductal Adenocarcinoma

Primary and secondary scientific-area codes, if applicable

Up to 80 chars

2 CTR; 4 IE

Project Summary / Abstract (approximately 30 lines)

Up to 3300 chars

This project proposes a Phase 1, first-in-human, open-label study of perioperative daraxonrasib (RMC-6236) with an eight-stage robotic pancreaticoduodenectomy for up to 18 adults with resectable or borderline-resectable KRAS G12-mutated pancreatic ductal adenocarcinoma. Daraxonrasib will be evaluated by 3+3 dose escalation at 160, 220, and 300 mg once daily [Kawchak, 2026a; Kawchak, 2026c].

The on-premises LLM is a bounded advisory system, not an autonomous controller. It may synchronize trial-document drafts, propose task-level surgical advisories from verified sensor context, and recommend a postoperative daraxonrasib restart window from protocol-defined inputs. Outputs are source-, model-, commit-, and seed-pinned; subjected to verification, validation, and uncertainty-quantification gates; logged; and accepted only after qualified human review. The LLM cannot authorize treatment or actuate the robot.

AI peer review will occur during development, not only after completion: ChatGPT Codex and Google Gemini will independently review Claude Code work at defined milestones, recommend corrections, and document disagreements for accountable human resolution and approval [Kawchak, 2025g; Kawchak, 2026g].

Primary endpoints are 30-day device/procedure serious adverse events, daraxonrasib MTD/RP2D, and completion of assigned tasks without unsafe conversion. Secondary outcomes include R0 resection, major fistula or complications, and 30- and 90-day mortality; exploratory outcomes include progression-free and overall survival, LLM concordance, and the simulation-to-real gap. Layered Sponsor-Investigator, IRB, DSMB, procedure-monitor, clinical-monitor, and Physical AI safety oversight will test whether verified generation can accelerate trial preparation and support surgery without weakening participant protection, human accountability, or data integrity [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026f; Kawchak, 2026g].

Project Narrative / Public Health Relevance (three plain-language sentences)

Up to 900 chars

Pancreatic cancer surgery is difficult, and patients need safer ways to evaluate new targeted therapies and surgical technologies. This study tests daraxonrasib with a human-controlled robotic Whipple in which an on-premises LLM may propose verified, logged advice but cannot authorize treatment or move the robot. The goal is to learn whether the combined drug-device approach is safe and feasible enough for later comparative testing.

Nontechnical overview for a foundation, patient organization, donor, or corporate review committee

Up to 1900 chars

This Phase 1 study is designed for up to 18 adults with KRAS G12-mutated pancreatic cancer that can be removed surgically. It combines perioperative daraxonrasib with an eight-arm robotic Whipple procedure and measures drug safety, surgical safety, task completion, and early oncologic outcomes [Kawchak, 2026a; Kawchak, 2026c].

The LLM is kept on premises and outside the robot's vendor control system. It can draft synchronized study materials, map verified sensor information into task-level surgical advice, and recommend when daraxonrasib might restart after surgery. It cannot approve a dose, choose to operate, calculate hidden vendor motion, or move an arm. Clinicians review each recommendation and can block, override, pause, convert, or stop the procedure.

Before any generated patient-facing recommendation is used, a verification suite tests source fidelity, timing, boundaries, uncertainty, and reproducibility. Accepted, blocked, escalated, and human-confirmed decisions are recorded in an audit trail. Funding supports the regulatory, clinical, engineering, data, and safety work needed to determine whether this combined approach can be studied responsibly, not to claim efficacy from a small early-phase trial [Kawchak, 2026f; Kawchak, 2026g].

One-line decision summary**In one sentence: what will the requested funding make possible that cannot otherwise happen?**

Up to 500 chars

Funding will enable a governed first-in-human evaluation of perioperative daraxonrasib with an LLM-advised robotic Whipple, including the regulatory, clinical, safety, and VVUQ infrastructure that simulation alone cannot establish [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026g].

3 Research Strategy / Pioneer-Style Essay (Next Page)

Agency-specific caution. RFA-RM-27-001 uses a five-page Research Strategy essay and instructs applicants not to submit Specific Aims or a detailed scientific plan. For another funder, rename, reorder, add, or delete the page modules to match that sponsor's instructions.

RESEARCH STRATEGY / ESSAY - PAGE 1 OF 5

Preparation note. Begin with the two scientific-area designations and a descriptive project title. Explain the major challenge or opportunity, importance, and broad scientific vision for a multidisciplinary audience.

Scientific Areas (primary; secondary)

Up to 80 chars

2 CTR; 4 IE

Project Title

Up to 220 chars

Daraxonrasib Phase 1 LLM-Advised Robotic Whipple Trial in KRAS-Mutated PDAC

Project description - major challenge or opportunity and why it matters

Up to 5700 chars

Pancreatic ductal adenocarcinoma (PDAC) concentrates biological, surgical, and translational risk. Even when resection is possible, pancreaticoduodenectomy is complex, perioperative complications may delay systemic therapy, and recurrence remains common. This project addresses resectable or borderline-resectable KRAS G12-mutated PDAC through a Phase 1 combination of perioperative daraxonrasib and an eight-arm robotic Whipple, evaluated in up to 18 treated participants by 3+3 dose escalation across 160, 220, and 300 mg once daily [[Kawchak, 2026a](#); [Kawchak, 2026c](#)].

The opportunity is not to attach generative AI to an existing operation. It is to test a clinical-trial architecture in which molecular therapy, robotic execution, human authority, Combined IND/IDE obligations, and computational assurance are specified together. The operation is staged from dissection through three anastomoses and final imaging/drain placement; arm duties and hard force, timing, watchdog, parking, and emergency-stop limits are prespecified and remain independent of the LLM.

The model has three bounded generation roles. First, it prepares synchronized drafts of protocol, consent, regulatory, safety-review, and operating artifacts from controlled sources. Second, during the procedure it maps verified sensor context into task-level Cartesian advisories; these are recommendations to the human team, not vendor kinematics or actuator commands. Third, after surgery it proposes a daraxonrasib pause-and-restart day using the protocol's fistula-grade and serum-trough inputs. Every output is pinned to source, model, repository commit, and seed; screened before generation; recorded in a hash-chained audit trail; and accepted only after qualified human confirmation [[Kawchak, 2026b](#); [Kawchak, 2026e](#); [Kawchak, 2026f](#); [Kawchak, 2026g](#)].

Improvements over time on coding, scientific-reasoning, and multimodal benchmarks support current frontier models as independent reviewers, but not as clinical validators. Cross-model review is therefore scheduled during development to identify disagreements, code defects, unsupported claims, and regulatory gaps before release [[Kawchak, 2025g](#); [Kawchak, 2025f](#)].

The Phase 1 study asks whether this governed drug-device-generation system can establish safety, feasibility, dose readiness, and an interpretable simulation-to-real gap. Primary outcomes are 30-day device/procedure serious adverse events, daraxonrasib MTD/RP2D, and completion of assigned tasks without unsafe conversion. Secondary outcomes include R0 resection, ISGPS grade B/C fistula, Clavien-Dindo grade III or higher complications, and 30- and 90-day mortality; PFS/OS through 24 months and LLM concordance are exploratory. Success would define a reproducible pathway for other Physical AI oncology trials. Failure would still identify boundaries, stopping criteria, and assurance requirements that the field currently lacks.

RESEARCH STRATEGY / ESSAY - PAGE 2 OF 5

Preparation note. Describe the pioneering approach and potential for groundbreaking or paradigm-shifting results. Convey rigor and robustness without presenting a conventional step-by-step protocol. Essential citations and figures, if used, consume space within the five-page limit.

Project description - innovative vision, conceptual framework, rigor, and potential impact

Up to 6200 chars

The pioneering concept is a verification-first clinical system in which generation is useful only inside explicit authority and assurance boundaries. The architecture links five layers: participant and disease state; perioperative daraxonrasib; the staged eight-arm Whipple; human clinical control; and an on-premises trust layer that evaluates every generated artifact or advisory. Drug, device, software, data, and oversight are co-designed rather than handed off among separate programs [Kawchak, 2026a; Kawchak, 2026c].

The intraoperative loop begins with 640 sensor channels: force channels sampled at 100 kHz and other channels at 10 kHz. Sensor context is mapped to a Cartesian task frame, and the repository-pinned LLM may issue a task-level advisory. A separate safety gate enforces vessel zones, per-arm and cumulative force caps, heartbeat and watchdog timing, and emergency-stop behavior before the vendor stack can act. The LLM is deliberately outside that stack. Its role is to offer a traceable second opinion; it has no independent authority over kinematics, actuation, conversion, drug administration, or patient management. Verification before generation supplies the rigor. A proposed interaction passes a ten-gate assurance suite aligned in the Phase 1 Protocol to external safety, robotics, and clinical baselines. Epistemic and aleatory uncertainty are made visible. The gate returns ACCEPT, BLOCK, or ESCALATE, and the human decision is logged with the exact source set, model/version, code commit, seed, inputs, outputs, and correction. Preclinical activation therefore requires fixed context of use, prohibited actions, interface limits, dry-run simulation in at least two frameworks, deterministic replay, boundary and timing tests, emergency-stop tests, and review of the residual-risk matrix [Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h].

The same rules govern document and dosing generations. Repository-based agents may prepare synchronized drafts across the clinical, regulatory, and operational package, but accountable authors verify every claim and approve every release. The perioperative advisor may propose restart at T+7, T+14, or T+21/continued hold from protocol-defined fistula grade and serum trough, but the treating team confirms or rejects the recommendation. These linked use cases test whether one provenance and governance discipline can cover textual, pharmacologic, and cyber-physical outputs without pretending that they have equal risks.

Clinical interpretation remains appropriate for a small Phase 1 study. The 3+3 design is not powered for efficacy. Safety, dose, surgical quality, and task-feasibility outcomes will be summarized descriptively with exact uncertainty intervals and prespecified evaluability rules. A Sponsor-Investigator, reviewing IRB, DSMB, Physical AI Safety Review Committee, independent procedure monitor, qualified operators, and robotics/AI-competent clinical monitoring provide layered review. Any treatment-related death, essential safety-override failure, uncontrolled cyber-physical event, or unacceptable toxicity can pause enrollment or require redesign.

The potential impact is broader than one drug or one operation. A successful study would show how generative systems can support regulated clinical work while remaining subordinate to verified evidence and human judgment. It would also produce reusable Combined IND/IDE logic, role definitions, audit structures, assurance gates, and simulation-to-clinical transition criteria. An unfavorable result would be equally informative because the system is designed to identify exactly where generation, sensing, robotic execution, or governance ceases to be acceptable.

RESEARCH STRATEGY / ESSAY - PAGE 3 OF 5

Preparation note. Provide concrete evidence of the PD/PI's innovativeness, such as integrating disparate information, questioning paradigms, working across disciplines, handling uncertainty, persisting through failure, and executing ambitious strategies.

Evidence of PD/PI innovativeness and AI-development provenance

Up to 6200 chars

Kevin Kawchak's recent work demonstrates an iterative LLM-assisted path from literature synthesis to a fundable clinical concept. In June 2025, an OpenAI deep-research workflow produced 40 PDAC meta-analyses totaling more than 400,000 words; Gemini subsequently ranked a daraxonrasib-containing combination as the leading candidate for a funding-oriented trial proposal [Kawchak, 2025n].

In August 2025, the associated QSP simulation story record reported median overall survival of 12.8 months for the daraxonrasib combination and 5.4 months for chemotherapy. In May 2026, the RASolute 302 trial reported median overall survival of 13.2 months with daraxonrasib and 6.6 months with chemotherapy in the RAS G12 population. The numerical proximity is an important chronology and hypothesis-supporting observation, not a validation claim, because the simulation and clinical trial differed in sample size, regimen, molecular selection, and evidentiary status [Kawchak, 2025a; O'Reilly et al., 2026].

In 2026, Claude Code was used to support extensive FDA, CFR, and ICH regulatory adaptations, followed by generation and revision of pancreatic-cancer protocols and an IND package. That sequence converted prior literature and simulation work into a sufficiently organized regulatory and clinical-document foundation to make the present funding application viable [Kawchak, 2026j; Kawchak, 2026k; Kawchak, 2026l; Kawchak, 2026b; Kawchak, 2026d; Kawchak, 2026a].

The planned tripartisan workflow assigns Claude Code the primary production role for code, regulatory adaptations, protocols, IND materials, diagrams, and repository-scale artifacts. ChatGPT Codex serves as the first independent peer reviewer, emphasizing verification, validation, uncertainty quantification, external standards, trial-literature synthesis, deep research, and PDF review. Google Gemini serves as the second independent reviewer, emphasizing accelerated code problem solving, meta-verification of prior stages, and administrative workflow support. Reviews occur at defined mid-project milestones and after material corrections so that findings can change the work before release, with the human PI retaining final authority [Kawchak, 2025g; Kawchak, 2026g; Kawchak, 2026e].

Each version of the current NIH application was primarily generated by ChatGPT, with Gemini providing additional fixes. These systems are disclosed as drafting and peer-review aids; they are not applicants, investigators, sponsors, regulators, clinicians, or decision-makers. The PI remains responsible for source verification, regulatory accuracy, scientific interpretation, participant protection, and final submission.

RESEARCH STRATEGY / ESSAY - PAGE 4 OF 5

Preparation note. Explain why the proposed work is a fundamental new direction rather than an obvious extension or scale-up of current or past work. Then explain why a high-risk/high-reward or otherwise special mechanism is a better fit than a conventional award.

How the planned research substantially differs from past or current work

Up to 3400 chars

The planned research is a fundamental new direction because the investigator's prior portfolio is predominantly computational, documentary, regulatory-conceptual, or simulation-based. It includes PDAC digital-twin proposals, large in silico trials, QSP models, VVUQ test frameworks, robotic simulations, protocol drafts, and AI-advised sponsor and site architectures [Kawchak, 2025e; Kawchak, 2025b; Kawchak, 2025a; Kawchak, 2026g; Kawchak, 2026q; Kawchak, 2026m]. Those works establish hypotheses, workflows, and assurance methods, but don't establish safety or feasibility in participants.

The proposed project crosses that boundary. It converts a body of simulated and document-based work into a prospective, first-in-human clinical investigation of perioperative daraxonrasib with an LLM-advised robotic Whipple in resectable or borderline-resectable KRAS G12-mutated PDAC [Kawchak, 2026a; Kawchak, 2026c]. This requires real regulatory authorization, qualified clinical sites, investigational product control, device accountability, surgeon and staff credentialing, informed consent, independent monitoring, validated data systems, and direct responsibility for participant welfare. The scientific question is no longer whether an AI system can generate a plausible protocol or complete a synthetic test suite. It is whether a bounded, human-governed version of the combined platform can meet prespecified clinical and cyber-physical safety requirements.

The work also differs from a conventional scale-up. It does not merely add patients, compute, prompts, or robotic arms to a prior simulation. The clinical context of use, risk classification, assurance case, endpoints, stopping boundaries, and human-machine authority structure must be redesigned for prospective use. The transition from simulated performance to accountable human-subject research is therefore the core innovation and the principal source of risk.

Suitability for the target program / why a traditional mechanism is insufficient

Up to 2300 chars

This project fits a high-risk, high-reward Pioneer mechanism because it proposes a potentially field changing clinical-trial architecture rather than an incremental optimization of an established procedure. The concept combines targeted KRAS therapy, complex pancreatic surgery, Physical AI, human oversight, regulatory harmonization, and VVUQ into a single prospective program [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026f]. The scientific and operational uncertainties are too interdependent for a narrowly scoped conventional award: the value of the drug-device concept cannot be separated from the assurance framework, site infrastructure, regulatory pathway, and real-time governance needed.

A traditional mechanism would also encourage fragmentation into sequential subprojects, increasing the risk that the clinical protocol, robotic system, data architecture, and safety case evolve incompatibly. The Pioneer structure supports a unified program with authority to stop, redesign, or redirect components as evidence emerges. It is especially appropriate because the most important deliverable may be a validated translational framework and a clear boundary on what should not proceed, rather than a predetermined positive efficacy result. The proposal aligns with the program's emphasis on innovative research, broad impact, and a substantial change in direction [NIH, 2026a].

RESEARCH STRATEGY / ESSAY - PAGE 5 OF 5

Preparation note. For the NIH Pioneer baseline, include the required effort commitment: more than six person-months (at least 51 percent) in years 1-3, at least four months (33 percent) in year 4, and at least three months (25 percent) in year 5. Explain adjustment of existing commitments when applicable.

Statement of research effort commitment and adjustment of existing commitments

Up to 2600 chars

If selected for award, Kevin Kawchak will commit more than 6 person-months of research effort, equivalent to at least 51 percent, in each of project years 1 through 3; at least 4 person-months, 33 percent, in project year 4; and at least 3 person-months, 25 percent, in project year 5, consistent with the Pioneer Award requirements [NIH, 2026a]. To make that commitment operationally credible, current non-sponsored and internally advised activities at ChemicalQDevice will be consolidated into the funded project, while routine website, general portfolio-maintenance, and nonessential exploratory drafting work will be reduced, deferred, or delegated. During the first three years, calendar priority will be assigned to regulatory filings, site-startup oversight, safety review, documentation governance, statistical review, and manuscript-ready curation of the trial platform; external talks, speculative side projects, and unrelated consulting will be capped or paused.

Repository-based workflows, versioned templates, and re-use of already established IND/protocol/ simulation assets will further reduce avoidable administrative overhead while preserving accountable human review [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2025d; Kawchak, 2026a; Kawchak, 2026c]. Based on the materials provided for this application draft, no conflicting active federal award requiring 6 or more person-months was identified; if any additional current or pending support exists at time of submission, the final other-support statement should be revised so that the exact source, overlap analysis, rebudgeting, delegation, no-cost extension, or relinquishment steps are stated explicitly. ChemicalQDevice will provide institutional confirmation that protected effort, essential decision authority, compute resources, and records stewardship needed for project are available across project period.

Concluding case for broad impact, feasibility, responsible risk management, and measurable success

Up to 3300 chars

This project seeks to establish whether a verification-governed combination of perioperative daraxonrasib and an LLM-advised robotic Whipple can enter early clinical testing without weakening the protections expected in oncology research. Its broad impact lies in the framework: a combined drug device protocol, explicit human authority, harmonized regulatory responsibilities, auditable provenance, and VVUQ gates that accept, escalate, or block patient-facing actions [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026f; Kawchak, 2026g].

Feasibility is supported by a coherent body of preparatory work spanning PDAC simulation, QSP and digital-twin methods, Phase 1 and Phase 2 protocol drafting, repository-based document workflows, federated data infrastructure, and standardized authorization and provenance services [Kawchak, 2025a; Kawchak, 2025c; Kawchak, 2026b; Kawchak, 2026d; Kawchak, 2026o; Kawchak, 2026n]. These products do not prove clinical performance; they reduce ambiguity about the questions, interfaces, documents, tests, and governance that must be resolved before enrollment. Risk will be managed through staged decision gates. During conduct, prespecified stopping rules will address drug toxicity, surgical complications, device failures, cyber-physical anomalies, protocol deviations, and unacceptable uncertainty. Human clinicians retain authority at every stage.

Measurable success will be defined by regulatory and site activation; completion of prespecified verification and validation tests; enrollment and retention against approved targets; interpretable estimates of safety and feasibility; complete auditability of patient-facing automated actions; timely safety review; and a final decision to proceed, redesign, or stop. The project is deliberately structured so that an unfavorable result remains scientifically valuable. By replacing unbounded enthusiasm for AI generation with prospective evidence and accountable control, the work could establish a durable standard for responsible Physical AI oncology trials.

4 Clinical Trial Synopsis

Preparation note. Complete this module when a clinical trial is proposed or when a funder requests a protocol synopsis. It is separate from the five-page Pioneer-style essay so that it can be included, omitted, or converted into the target funder's study-record format.

Protocol title

On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma [Kawchak, 2026c]

Protocol / version / date

Protocol v1.0.0, June 21, 2026

Phase

Phase 1 [Kawchak, 2026a; Kawchak, 2026c]

Study type

First-in-human, prospective, open-label, single-arm interventional clinical study [Kawchak, 2026a]

Condition / indication

Resectable or borderline-resectable KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC) [Kawchak, 2026a]

Investigational intervention

Perioperative daraxonrasib with a human-controlled eight-arm robotic Whipple and on-premises LLM advisory [Kawchak, 2026a; Kawchak, 2026c]

Comparator / control

None for the Phase 1 single-arm study; comparative evaluation is deferred to the later randomized Phase 2 study [Kawchak, 2026a; Kawchak, 2026d]

Allocation / masking

Phase I: Non-randomized, Open-label, Single-arm.

Target enrollment

Up to 18 treated participants

Number and type of sites

Single qualified pancreatic surgery site; final institution pending agreement

Study duration / follow-up

Up to 28-day screening; 30/90-day safety; 24-month OS follow-up

Scientific premise and clinical rationale

Up to 1400 chars

The Phase 1 Protocol combines perioperative daraxonrasib with a staged eight-arm Whipple in KRAS G12-mutated resectable or borderline-resectable PDAC. Its scientific premise is that targeted therapy and robotic execution can be evaluated as one governed investigational system only when the LLM remains advisory, the human team retains authority, and every generated artifact, surgical recommendation, and restart recommendation is verified and logged before use [Kawchak, 2026a; Kawchak, 2026c].

Primary objective, endpoint, estimand, and time point

Up to 1300 chars

Primary objective: establish the initial safety and feasibility of the combined daraxonrasib, robotic Whipple, and governed LLM-advisory system.

Co-primary endpoints are: (1) incidence of device- or procedure-related serious adverse events through 30 days after surgery; (2) daraxonrasib dose-limiting toxicities and determination of the maximum tolerated dose / recommended Phase 2 dose under the 3+3 design; and (3) completion of protocol-assigned robotic tasks without unsafe conversion. The third endpoint counts conversion ordered for patient safety as clinically appropriate while separately identifying conversion caused by unacceptable device, advisory, or integration failure.

Primary analyses use all relevant treated or procedure-evaluable participants, retain dose holds, rescue actions, and open conversion in safety summaries, and report counts, proportions, exact confidence intervals, timing, attribution, and cohort-level decisions. Ninety-day events provide supportive safety context [[Kawchak, 2026a](#); [Kawchak, 2026c](#)].

Key secondary and exploratory objectives / endpoints

Up to 1250 chars

Secondary surgical and oncologic endpoints are R0 resection rate; ISGPS grade B/C postoperative pancreatic fistula; Clavien-Dindo grade III or higher complications; and 30- and 90-day mortality. Additional descriptive measures include conversion, operative completion by P1-P8 phase, hospital stay, readmission, reoperation, and time to postoperative therapy readiness.

Exploratory endpoints include progression-free and overall survival through 24 months; LLM advisory concordance with the qualified human decision; ACCEPT/BLOCK/ESCALATE frequency; latency and emergency-stop performance; anastomotic tension and force-limit adherence; document-generation correction burden; and the simulation-to-real gap across sensor, timing, task, and safety outcomes. These endpoints are hypothesis-generating and do not support efficacy claims in the Phase 1 cohort [[Kawchak, 2026c](#); [Kawchak, 2026b](#)].

Eligibility criteria and target population

Up to 1400 chars

Target population: adults with resectable or borderline-resectable pancreatic ductal adenocarcinoma harboring a KRAS G12 alteration for whom curative-intent pancreaticoduodenectomy is clinically appropriate [[Kawchak, 2026a](#); [Kawchak, 2026c](#)]. Key inclusion criteria are age 18 years or older; histologic or cytologic evidence of PDAC or unequivocal clinicopathologic diagnosis per protocol; documented KRAS G12-mutated status from a CLIA-qualified or equivalently validated assay; ECOG performance status 0 to 1; adequate marrow, hepatic, renal, and cardiopulmonary function; resectable or borderline-resectable disease without distant metastasis; anatomy suitable for the planned robotic approach; and ability to understand and sign informed consent. Key exclusion criteria are metastatic disease, uncontrolled infection, clinically significant intercurrent illness, prior intolerance or contraindication to study drug or major pancreatic surgery, pregnancy or breastfeeding, inability to comply with perioperative monitoring, and any medical, technical, or psychosocial condition that, in the investigator's judgment, would make study participation unsafe or uninterpretable.

Phase 1 Protocol Figure 1 – Overall Trial Schema

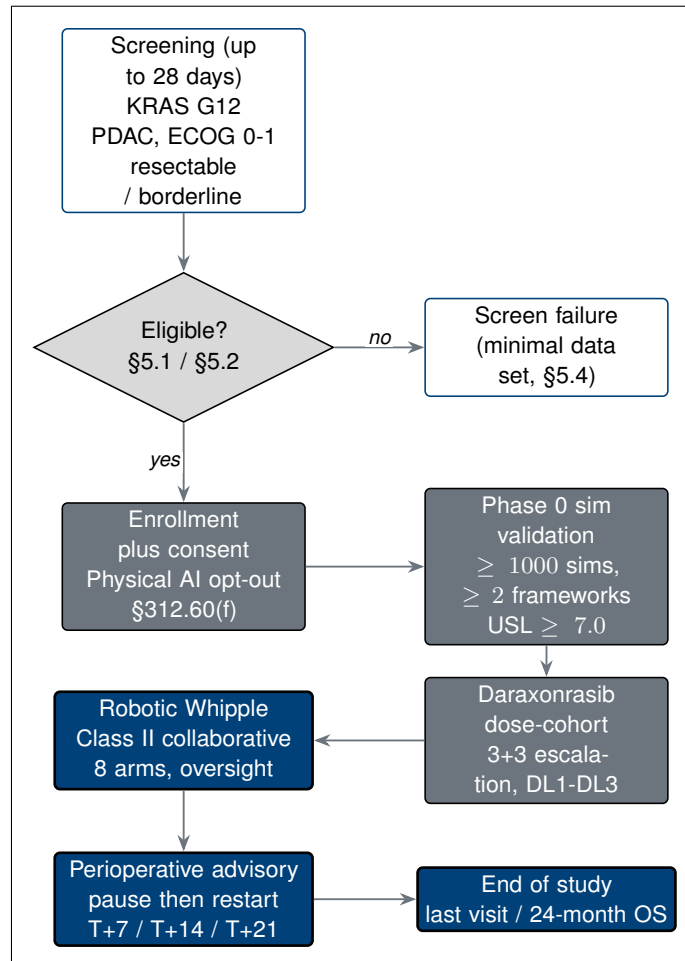


Figure 1. Overall trial schema for the Phase 1 combined IND/IDE study, from 28-day screening through enrollment with the Physical AI consent opt-out, Phase 0 simulation validation, daraxonrasib 3+3 dose assignment, the eight arm robotic Whipple, and perioperative pause-and-restart advisory. Robot USL = Unification Standard Level.

Phase 1 Protocol Figure 2 – Objective-to-Endpoint Hierarchy

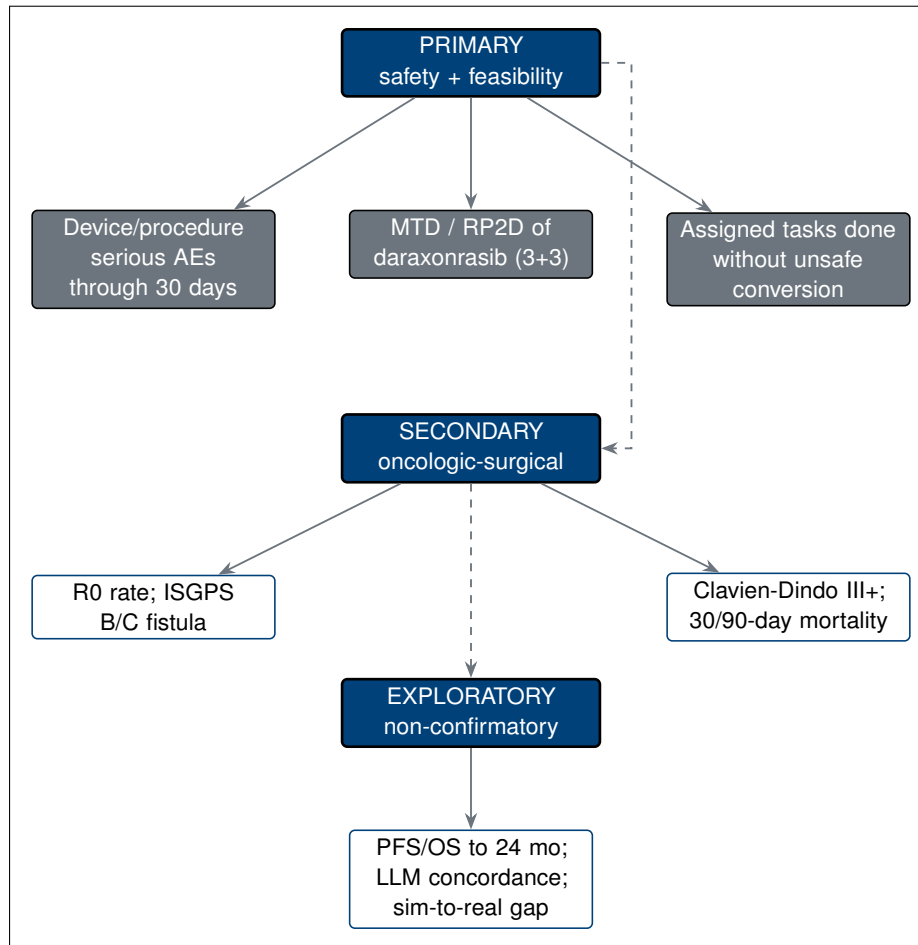


Figure 2. Objective-to-endpoint hierarchy. Primary safety and feasibility (30-day serious adverse event incidence, daraxonrasib MTD/RP2D, and unsafe-conversion-free task completion) sit above secondary oncologic-surgical quality (R0 rate, ISGPS grade B/C fistula, Clavien-Dindo grade III+, and 30- and 90-day mortality) and exploratory progression-free and overall survival, LLM advisory concordance, and the sim-to-real gap.

5 Human Subjects, Ethics, Inclusion, and Regulatory Strategy

Preparation note. Use this section as a drafting companion to the required human-subjects and clinical-trials forms. It does not replace an IRB protocol, informed-consent document, FDA submission, single-IRB plan, or agency-specific study record.

Human subjects involved?

Yes

NIH-defined clinical trial?

Yes

Delayed onset?

No

IND / IDE status

Combined IND/IDE; FDA status pending

IRB / sIRB status

IRB in preparation; sIRB N/A

ClinicalTrials.gov status

PRS package setup pending

Risk to participants and adequacy of protections

Up to 1800 chars

Risks include daraxonrasib toxicity; anesthesia and pancreaticoduodenectomy morbidity; bleeding, infection, fistula, delayed gastric emptying, readmission, and death; device malfunction; incorrect or delayed advisory output; cybersecurity events; privacy loss; and therapeutic misconception. The combined intervention is investigational and cannot be represented as superior to standard care [Kawchak, 2026a; Kawchak, 2026c].

Protections include multidisciplinary eligibility review, qualified pancreatic surgeons and operators, standard anesthesia and rescue resources, independent procedure monitoring, DSMB/IRB oversight, cohort holds, and explicit conversion or termination authority. The LLM is on premises, repository-pinned, and outside the vendor kinematic stack. It may draft study artifacts, issue task-level advisories from verified sensor context, and propose a postoperative restart day, but it cannot authorize treatment or actuate the robot. Every patient-facing output passes verification-before-generation gates and requires human confirmation. Hard force, timing, watchdog, park, and emergency-stop controls remain available without the LLM, and every decision is auditable.

Potential benefits, importance of knowledge, and risk-benefit assessment

Up to 1400 chars

Direct clinical benefit cannot be promised, and some participants may receive no therapeutic benefit beyond standard perioperative management. Potential direct benefits include tumor-control or operative advantages sufficient to support successful resection, improved perioperative coordination, and closer protocol-driven monitoring; however, these are hypotheses rather than established outcomes in this first-in-human setting [Kawchak, 2026a; Kawchak, 2026c]. The knowledge value is substantial because the study addresses whether a targeted therapy, robotic procedure, and verification-first AI governance layer can be introduced into oncology research under accountable regulatory, ethical, and operational controls. The risk-benefit assessment is therefore acceptable only with strict eligibility, staged activation, intensive oversight, rapid stopping rules, and transparent governance. In a disease with poor prognosis and limited curative opportunity, the proposed bounded Phase 1 design is justified as a safety and feasibility study, not as a claim of superior efficacy [Kawchak, 2026d; Kawchak, 2026f; NIH, 2026a].

Informed consent / assent, capacity, and additional safeguards for vulnerable populations

Up to 1650 chars

Informed consent will be obtained by qualified study personnel before any research-only procedure, using an IRB-approved consent form written in plain language and a process that gives prospective participants adequate time to review the study, ask questions, discuss alternatives, and decide free of coercion [Kawchak, 2026k]. Because the initial target population is competent adults undergoing high risk cancer surgery, assent is not expected to apply. Decisional capacity will be assessed clinically, and consent will not be obtained during acute sedation, severe pain crises, or other conditions that materially impair understanding. Key disclosures will include the first-in-human nature of the program, uncertain benefit, major surgical and drug risks, robotic and software governance features, alternatives to participation, privacy protections, ClinicalTrials.gov posting, and the right to withdraw. Additional safeguards include optional re-consent after material protocol change, witness procedures when needed, translated materials and interpreter support for participants with limited English proficiency, and exclusion of prisoners, children, and other populations requiring additional subpart-specific protections unless a future amendment supplies the required regulatory justifications and safeguards.

Inclusion by sex, race, ethnicity, and age; scientific justification for any exclusions

Up to 1750 chars

Enrollment will be open to eligible adults regardless of sex, gender, race, or ethnicity, and recruitment materials, screening logs, and accrual reviews will be used to monitor whether study participation reflects the catchment area served by the participating site [NIH, 2026a]. No race- or ethnicity-based exclusions are scientifically justified. The protocol should aim for broad inclusion of women and men and for equitable access among historically underrepresented populations through bilingual outreach, navigator support, flexible scheduling, and travel/logistics assistance when feasible. Age inclusion will begin at 18 years and older, with no arbitrary upper-age exclusion; older adults may participate if surgical candidacy and organ function criteria are met. Exclusion of children is scientifically and ethically justified because pancreaticoduodenectomy with perioperative daraxonrasib in KRAS G12-mutated PDAC is an adult first-in-human oncology intervention with disease epidemiology, operative risk, and dose/procedure uncertainty that do not support pediatric enrollment at this stage.

Privacy, confidentiality, HIPAA, Certificate of Confidentiality, and controlled-access strategy

Up to 1450 chars

Participant privacy and confidentiality will be protected through coded study identifiers, separation of direct identifiers from analytic datasets, minimum-necessary data access, role-based permissions, audit-trailed systems, and secure transfer and storage procedures. Research use and disclosure of protected health information will occur under IRB-approved HIPAA authorization or other legally permissible mechanism, with clear notice of what information will be used, who may receive it, and how participants may revoke authorization where applicable. If NIH funding is awarded and the project collects or uses identifiable, sensitive information, the study will be automatically deemed issued a Certificate of Confidentiality; non-NIH-funded linked studies would require separate CoC review if applicable [NIH, 2026a]. Only de-identified or suitably transformed datasets will be shared publicly; datasets with residual re-identification risk, operative video, detailed device logs, or linkable biospecimen/genomic variables will use controlled-access release, documented data-use conditions, and repository or enclave controls.

Regulatory pathway, required submissions, and responsible parties

Up to 1450 chars

The application anticipates a coordinated drug-device pathway: daraxonrasib under IND requirements and the significant-risk robotic system under an IDE determination, with FDA and IRB authorization required before clinical use. The Phase 1 Protocol's Combined IND/IDE diagram is a planning framework, not evidence that either authorization has been granted [Kawchak, 2026a; Kawchak, 2026c].

Phase 1 Protocol Figure 3 – Combined IND / IDE

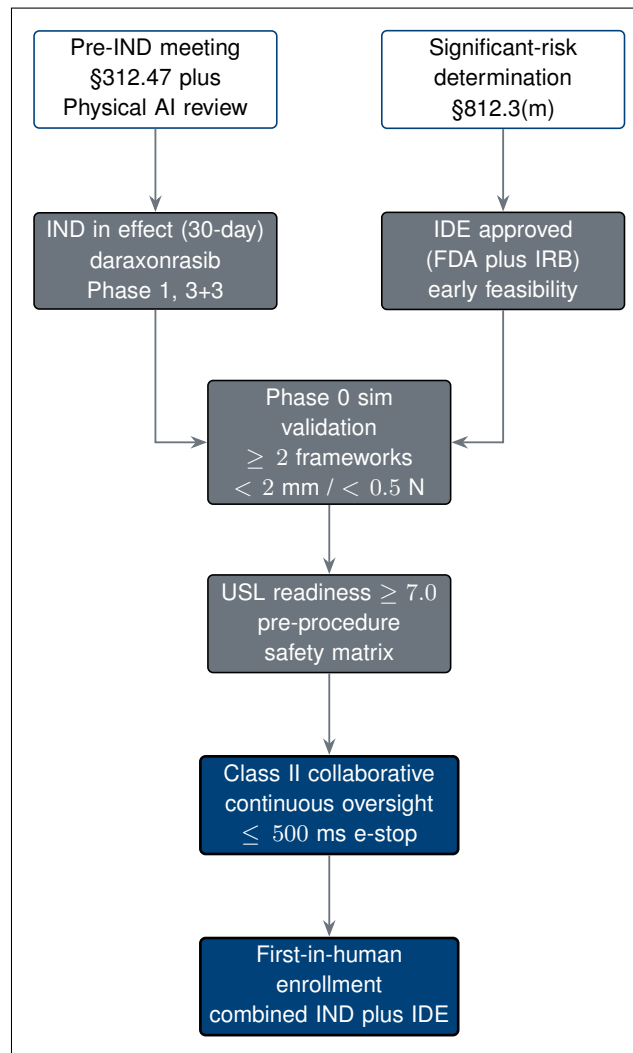


Figure 3. Combined IND / IDE regulatory pathway with the Physical AI overlay. The daraxonrasib drug arm (IND, part 312) and the robotic Whipple device arm (IDE, part 812, significant-risk per §812.3(m)) converge.

The Sponsor-Investigator owns regulatory submissions, safety reporting, protocol and software-change control, investigational product/device accountability, ClinicalTrials.gov registration/results reporting, and maintenance of the trial master file. The site may not enroll until the final FDA/IRB status, approved device/software configuration, model and commit hashes, consent, monitoring plan, and emergency procedures are documented. Material changes to LLM context of use, generation boundaries, safety gates, kinematics interfaces, or restart logic require prospective impact assessment and any FDA/IRB review.

6 Statistical Design, Safety Monitoring, Recruitment, and Operations

Preparation note. The NIH clinical-trial review criteria emphasize justified endpoints, interpretable design, adequate power, efficient conduct, ethical considerations, recruitment and retention, randomization or masking when appropriate, data management, safety monitoring, and feasible timelines.

Study design and analysis framework, including estimands and handling of intercurrent events

Up to 1700 chars

This is a first-in-human, prospective, open-label, single-arm Phase 1 study using a conventional 3+3 daraxonrasib escalation across 160, 220, and 300 mg once daily, with up to 18 treated participants [Kawchak, 2026a; Kawchak, 2026c]. Cohort progression considers dose-limiting toxicity, 30-day device/procedure serious adverse events, completion of assigned robotic tasks without unsafe conversion, protocol deviations, and cumulative cyber-physical findings.

Analyses are descriptive and decision-oriented. Dose holds, reductions, perioperative interruption, rescue actions, and conversion remain in the treatment-strategy safety summaries. Participants reaching the operating room contribute to procedure analyses; DLT evaluability follows protocol exposure rules, with an early DLT remaining evaluable. Counts, proportions, exact confidence intervals, times to event, and participant-level listings will be reported. Secondary and exploratory outcomes are not multiplicity-controlled and will be labeled hypothesis-generating.

Sample-size justification, assumptions, power or precision, and simulation strategy

Up to 1550 chars

The maximum sample is 18 treated participants because three dose levels are evaluated with a 3+3 escalation, with expansion from three to six participants at a level when protocol-defined DLT rules require it [Kawchak, 2026a]. The design is not powered to compare efficacy or to estimate uncommon risks precisely. Its purpose is to identify an initial MTD/RP2D, detect obvious unacceptable toxicity or procedural failure, and determine whether the assigned robotic tasks can be completed under the specified human-control and assurance boundaries.

Exact binomial intervals and cohort-level event patterns will communicate uncertainty. Phase 0 simulations in at least two frameworks stress-test timing, force limits, task allocation, restart rules, missingness, and stopping scenarios, but simulated patients and synthetic hardware cannot replace observed clinical data. The simulation-to-real gap is therefore an explicit exploratory outcome rather than an assumed bridge to efficacy.

Randomization, allocation concealment, masking, controls, and unblinding

Up to 1450 chars

No randomization is used because this is a first-in-human, single-arm Phase 1 study intended to estimate safety, tolerability, and feasibility rather than comparative efficacy [Kawchak, 2026a; Kawchak, 2026c]. Accordingly, allocation concealment is not applicable, the study is open label, and formal participant or investigator masking is not practical for the combined perioperative drug and robotic procedure. There is no concurrent control arm; external benchmarks and the later randomized Phase 2 protocol inform interpretation but do not function as statistical controls in this study [Kawchak, 2026d]. To reduce bias despite the absence of blinding, key operational and safety triggers will be prespecified, adverse-event coding will follow standard terminology, and independent review by the medical monitor or safety committee will be used for dose-escalation or continuation decisions. Unblinding rules are not applicable beyond restricted access to protected source records.

Primary and secondary analysis populations, missing data, multiplicity, and sensitivity analyses

Up to 1700 chars

The primary safety population will include all enrolled participants who receive any study daraxonrasib exposure. A procedure population will include participants who proceed to the intended operative encounter, and a treated-protocol or DLT-evaluable population will include participants with sufficient exposure to contribute meaningfully to the primary assessment window unless an early event itself defines evaluability. Secondary surgical and biomarker analyses will use endpoint-specific evaluable sets that are defined prospectively in the SAP [Kawchak, 2026a; Kawchak, 2026c]. Missing data will be minimized through near-real-time source review, remote follow-up options, and prespecified window management; safety endpoints will prioritize conservative ascertainment, while descriptive secondary endpoints will report missingness transparently rather than relying on unjustified imputation. No formal multiplicity adjustment is planned for secondary or exploratory endpoints because they are descriptive and hypothesis-generating. Sensitivity analyses will examine the impact of replacements, open conversion cases, early withdrawal, and endpoint-window definitions.

Recruitment, outreach, enrollment, retention, missed visits, and loss-to-follow-up mitigation

Up to 1800 chars

Recruitment will focus on a high-volume pancreatic oncology catchment through multidisciplinary tumor board review, surgical oncology and medical oncology referral pathways, molecular prescreening for KRAS G12 status, and rapid pre-eligibility confirmation prior to consent [Kawchak, 2026a; Kawchak, 2026c]. Outreach materials will be plain-language, culturally attentive, and available in translated form when needed. Enrollment feasibility will be supported by prespecified screening logs, weekly startup huddles during active recruitment, and replacement rules for screen failures or pre-treatment withdrawal. Retention will be strengthened through participant navigation, visit coordination around standard-of-care appointments, reminder systems, travel or lodging support when feasible, and remote or local-lab follow-up options for appropriate post-discharge assessments. Missed visits will trigger same-week contact attempts, windowed rescheduling, investigator review of clinically critical assessments, and use of routine care data when permitted. Loss to follow-up mitigation includes multiple contact methods, designated backup contacts, proactive scheduling of postoperative visits before discharge, and rapid outreach after any missed safety assessment.

Data and safety monitoring plan; DSMB / medical monitor; stopping and escalation rules

Up to 1800 chars

The Sponsor-Investigator is accountable to the reviewing IRB and DSMB and convenes a Physical AI Safety Review Committee at least every 90 days. An Independent Safety Monitor is designated for each procedure, and clinical monitoring must include robotics and AI competence. The site team includes the surgeon, primary operator, and qualified backup operator [Kawchak, 2026a; Kawchak, 2026c].

Safety review integrates daraxonrasib DLTs, serious adverse events, ISGPS and Clavien-Dindo complications, device malfunctions, force/timing violations, emergency-stop events, blocked or escalated advisories, and audit-trail integrity. Enrollment pauses for a treatment-related death, an essential override failure, an uncontrolled cyber-physical event, an unexpected severe-toxicity cluster, or direction from FDA, IRB, DSMB, or the Sponsor-Investigator. Review is required before cohort progression, restart after a hold, protocol modification, or device/software release.

Site activation, training, quality management, monitoring, auditing, and corrective actions

Up to 1550 chars

Activation requires regulatory and IRB clearance, frozen protocol and context of use, trained and credentialed personnel, drug/device accountability, validated data flow, and successful dry runs of P1-P8 tasks and emergency procedures. The exact LLM source set, model/version, repository commit, seed, prompts, gate results, human confirmations, and corrections are retained for every patient-facing generation.

Quality controls verify isolation from the vendor kinematic stack; role-based authorization; per-arm and cumulative force limits; heartbeat, watchdog, park, and emergency-stop timing; deterministic replay; and source-to-eCRF reconciliation. Risk-based monitoring prioritizes eligibility, consent, dosing, assigned task completion, serious events, conversion decisions, advisory concordance, and audit-chain continuity. A major finding triggers root-cause analysis, CAPA, retraining, and re-verification before unrestricted continuation.

Operational timeline, enrollment rate, milestones, contingencies, and decision gates

Up to 1550 chars

Months 0–6: finalize Combined IND/IDE strategy, IRB and ClinicalTrials.gov records, contracts, site qualification, data systems, assurance-case review, and Phase 0 simulation readiness. Months 7–30: enroll sequential 3+3 cohorts, up to 18 treated participants, with documented cohort reviews before escalation or continuation. Thirty- and 90-day safety follow-up continues after each procedure; progression-free and overall survival are followed through 24 months. Years 3–4: complete follow-up, data cleaning, endpoint adjudication, database lock, and the integrated Phase 1 decision. Year 5: publish results, release permitted data/code/provenance artifacts, and prepare a redesigned or later-stage program only if the safety, dose, task-feasibility, auditability, and simulation-to-real criteria support it. Slow accrual, hardware downtime, advisory-gate instability, or regulatory questions trigger predefined referral, repair, revalidation, or hold contingencies.

Phase 1 Protocol Figure 4 Advisory Commands, Figure 5 Daraxonrasib Restart

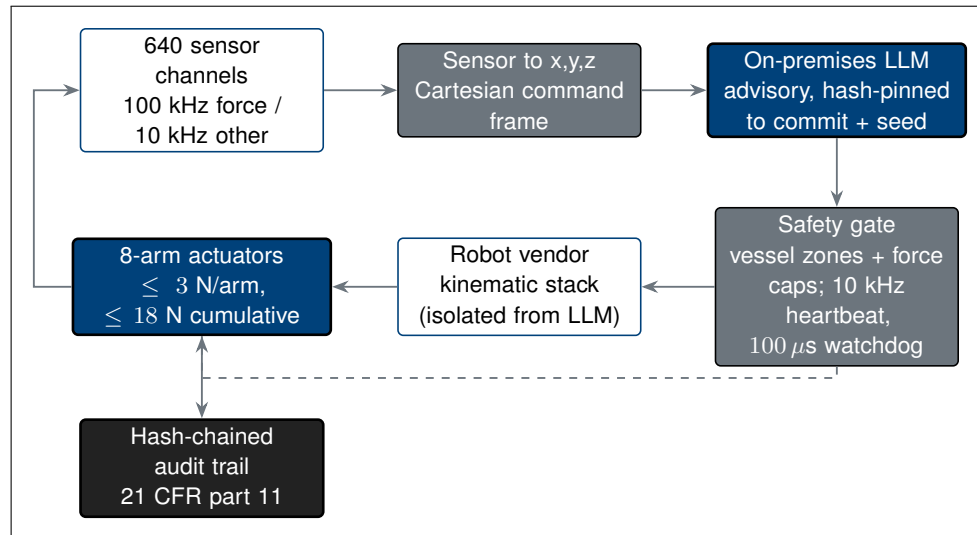


Figure 4. LLM issues hash-pinned advisory commands from outside the vendor kinematic stack (second-opinion oracle isolation); and a safety gate enforces the vessel zones, force caps, 10 kHz heartbeat, 100 μs watchdog, and ≤ 3 ms E-stop before actuator moves. Actuator telemetry returns to stack; steps written to 21 CFR part 11 trail.

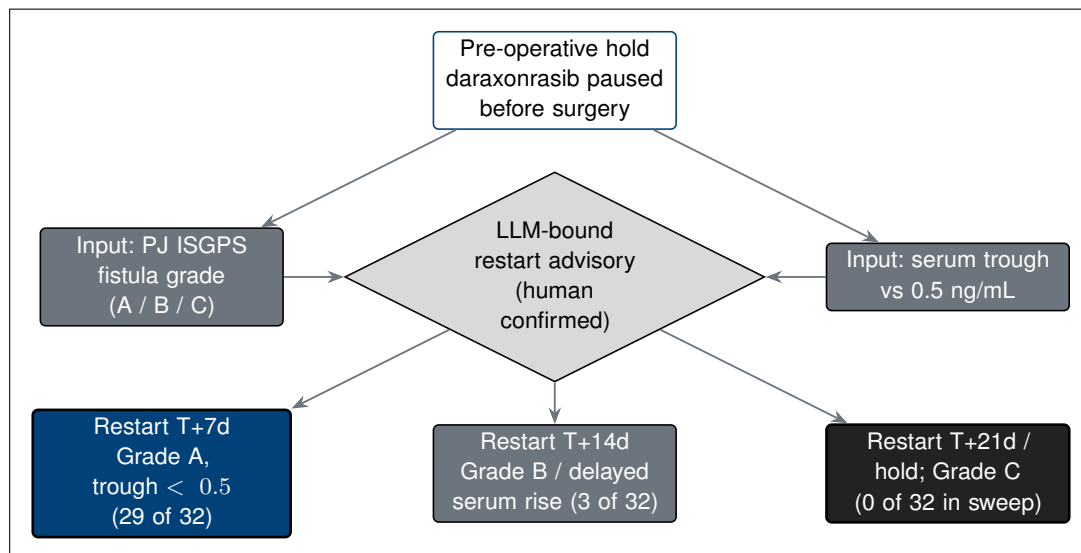


Figure 5. Daraxonrasib restart day is keyed to the pancreaticojejunostomy ISGPS fistula grade and serum trough relative to 0.5 ng/mL: T+7d for Grade A with a sub threshold trough (29 of 32 simulation iterations), T+14d for Grade B or a delayed serum rise (3 of 32), and T+21d or continued hold for Grade C (0 of 32 in the sweep).

Phase 1 Protocol Figure 6 – Study Governance, Figure 7 – Verification Before Generation

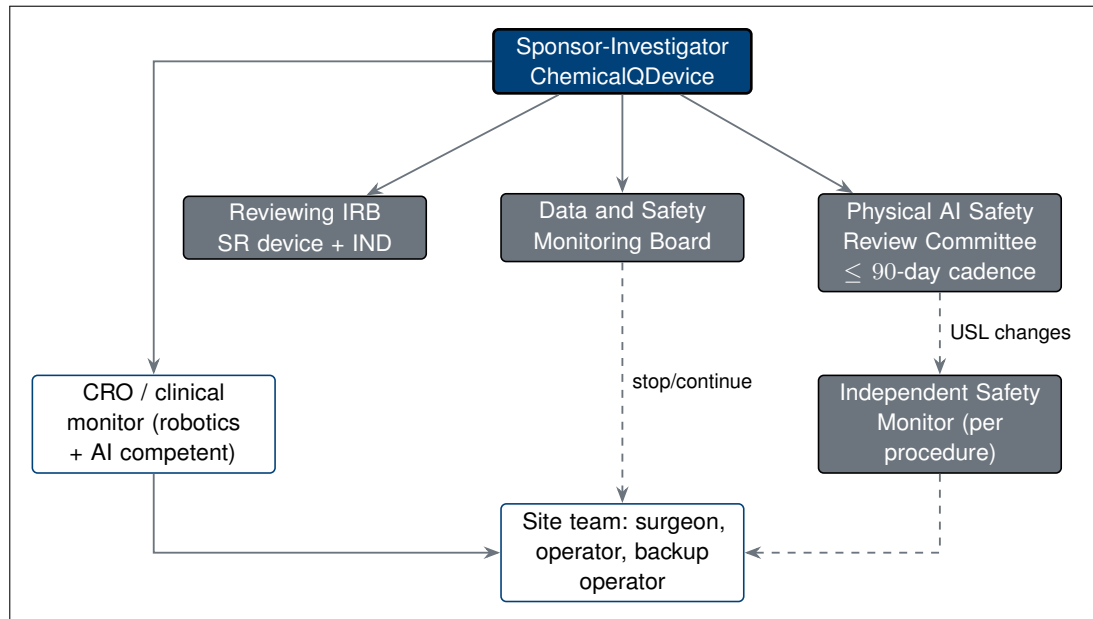


Figure 6. Sponsor-Investigator answers to the reviewing IRB and the DSMB, convenes a Physical AI Safety Review Committee on a ≤ 90 -day cadence, designates an Independent Safety Monitor for each procedure, and delegates monitoring to a robotics- and AI-competent CRO over site team (surgeon, operator, and a qualified backup operator).

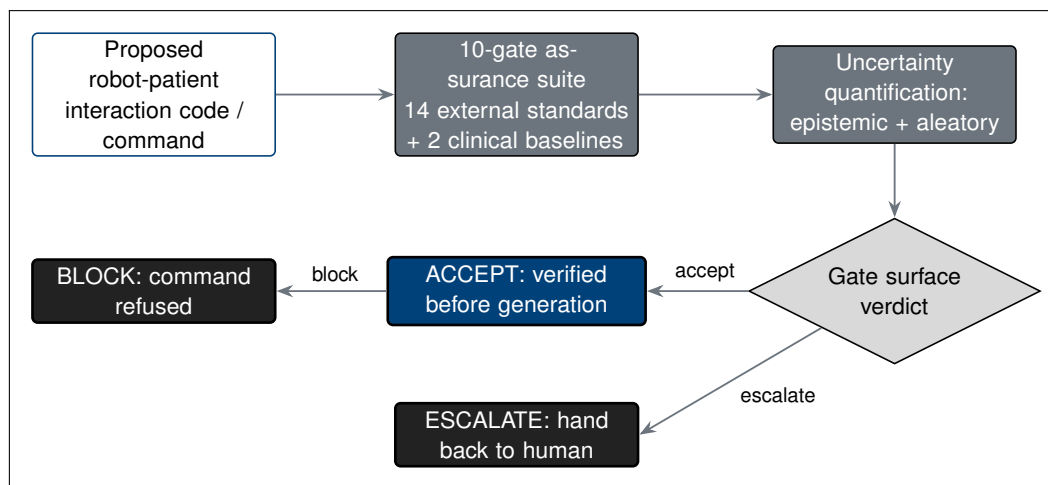


Figure 7. Verification before generation. Every proposed robot-patient command passes a ten-gate assurance suite aligned to fourteen external safety and robotics standards plus two clinical baselines, with epistemic and aleatory uncertainty bounds, before the gate surface returns ACCEPT, BLOCK, or ESCALATE (hand back to human).

7 Data Management, Sharing, Repositories, and Cybersecurity

Preparation note. The 2026 NIH pilot DMS format uses structured yes/no questions, a 300-word limitation rationale when sharing is restricted, and a short table of anticipated scientific data types and repositories. Adapt this module when another sponsor uses different data-sharing terms.

Maximum appropriate sharing of data underlying publications and other findings?

Shared by publication or, for other findings, by end of performance period?

Available for at least the period required by repository and/or journal policy?

Human-participant privacy, rights, and confidentiality protected, including access controls?

Subject to NIH Genomic Data Sharing policy?

Limitations on sharing and ethical, legal, or technical rationale (300 words maximum for NIH baseline)

Up to 2400 chars

The project will maximize sharing of de-identified scientific data, protocols, analysis code, data dictionaries, and supporting metadata consistent with participant safety and law. Open public release will not be used for direct identifiers, dates or combinations of indirect identifiers that create more than a very small re-identification risk, raw operative video containing recognizable anatomy or staff audio, detailed device telemetry that could reveal institutional cybersecurity architecture, or data whose release would conflict with consent, HIPAA, FDA obligations, or contractual restrictions on third-party software. Such materials, if shared at all, will use controlled-access repositories, curated de-identification, data-use terms, and documented review of requester qualifications and use purpose. If future NIH-funded genomic sequencing is added by amendment, the DMS plan will be updated to address the Genomic Data Sharing policy and the appropriate controlled-access repository. These limits are not intended to impede reproducibility; rather, they preserve participant rights while allowing broad release of the data needed to validate reported conclusions.

Expected scientific data and repositories

Data type / species / modality	Repository or example repository	Access
De-identified protocol, results, recruitment status, metadata	ClinicalTrials.gov	Public
De-identified clinical, safety, operative, case report form datasets	Controlled-access repository or secure data enclave	Controlled
De-identified radiology, pathology, imaging files, metadata	NCI Imaging Data Commons or The Cancer Imaging Archive	Open
Analysis code, SAP, simulation code, data dictionaries	GitHub with Zenodo DOI archiving	Public

Data standards, metadata, provenance, versioning, quality control, and reproducibility

Up to 1400 chars

Clinical data will use protocol-defined variables, controlled terminology, standard adverse-event and surgical classifications, and machine-readable dictionaries. Operative data include P1-P8 task identifiers, arm assignment, force/tension, timing, conversion, and emergency-stop events. Generated-artifact records include source identifiers, model/version, repository commit, seed, prompt or command context, ten-gate results, uncertainty outputs, ACCEPT/BLOCK/ESCALATE disposition, human confirmation, correction, and final release. Versioned schemas, checksums, immutable timestamps, audit trails, data-validation rules, and source-to-eCRF reconciliation preserve provenance. DICOM and FHIR-compatible structures will be used where suitable, without forcing local systems to expose unnecessary identifiers. Analysis code, metadata, and permitted derived data will be released with persistent versions and documentation sufficient for deterministic replay where confidentiality and cybersecurity allow [NIH, 2026c; Kawchak, 2026c].

Security architecture, role-based access, encryption, incident response, backup, and disaster recovery

Up to 1600 chars

Identifiable clinical data, raw operative media, detailed telemetry, model inputs/outputs, and regulatory records will remain in approved encrypted systems with least-privilege role-based access, multifactor authentication, environment separation, logging, backups, and tested recovery. The on-premises LLM has no unrestricted internet access during patient-facing use and is isolated from the vendor kinematic stack.

Security monitoring covers account changes, model or code substitution, unexpected network paths, audit-chain breaks, timing anomalies, unauthorized export, and device-control boundary violations. Each released model/software configuration is hash-pinned and independently verified before use. An incident can block generation, trigger human handback, stop an arm or the entire procedure through independent hardware controls, pause enrollment, preserve evidence, and initiate institutional, IRB, FDA, participant, and legal notifications as required. Public sharing excludes configurations that would materially increase re-identification or cyber-physical risk.

Governance, stewardship roles, costs, repository fees, and plan-update process

Up to 1350 chars

The sponsor-investigator will hold overall stewardship responsibility, with delegated roles for data management, biostatistics, ClinicalTrials.gov submission, repository preparation, cybersecurity, and records retention. A named data steward will maintain the release inventory, confirm de-identification and metadata completeness, and coordinate repository submissions at publication and project close. Budget planning should explicitly reserve funds for EDC hosting, secure storage, repository deposit or curation fees if incurred, long-term archival storage, and staff time for QC, documentation, and access-review workflows. The DMS plan will be reviewed at least annually and whenever a material protocol change occurs, particularly if new imaging, biospecimen, genomic, or device-log data types are added [NIH, 2026a].

8 Facilities and Other Resources

Agency-specific caution. The RFA-RM-27-001 baseline allows a one-page Facilities and Other Resources statement and says not to use a separate Equipment attachment. Keep the content on this page for the NIH baseline; modify the module for another sponsor.

Scientific environment, institutional support, and unique features

Up to 2400 chars

The project requires a qualified pancreatic cancer clinical environment rather than a legislative or purely virtual setting. Activation depends on documented access to a high-volume pancreatic surgery program, regulated research pharmacy, pathology and radiology, perioperative critical care, emergency conversion capability, secure on-premises compute, simulation space, device maintenance, and clinical research operations. ChemicalQDevice supplies the protocol, repository, assurance, and records-governance framework; site capabilities must be confirmed by formal agreement before enrollment.

Clinical sites, subject populations, laboratories, pharmacies, imaging, testing, and specimen resources

Up to 2300 chars

A final site must provide multidisciplinary pancreatic oncology review; molecular confirmation of KRAS G12 status; operating-room access for complex robotic pancreaticoduodenectomy with open-conversion readiness; anesthesia, blood bank, ICU, interventional radiology, endoscopy, and surgical rescue; investigational pharmacy; CLIA-qualified laboratory support; pathology margin assessment; NIR/ICG imaging; and 30-day, 90-day, and 24-month follow-up. No NCI designation or patient-volume claim should be made until the selected institution documents it.

Data coordinating, biostatistics, informatics, EDC, cybersecurity, and records infrastructure

Up to 1900 chars

Required resources include validated EDC, role-based source access, secure imaging and operative-media storage, high-frequency telemetry capture, on-premises LLM inference, version-controlled repositories, hash and time-stamp services, protected backups, statistical programming, and an audit environment that can reconstruct every generated advisory and human decision. The local network must enforce separation between the LLM advisory layer, the vendor kinematic stack, research systems, and public networks.

Collaborative arrangements and access to specialized capabilities

Up to 1700 chars

Before activation, executed arrangements are needed for pancreatic surgical oncology, medical oncology, anesthesia, pharmacy, pathology, radiology, biostatistics, data management, independent medical monitoring, DSMB participation, Physical AI safety review, robotics/device engineering, cybersecurity, and regulatory monitoring. The application identifies these functions without implying that unconfirmed individuals or institutions are committed. The NIH profile remains single-PI; collaborators are described where RFA permits.

Institutional commitments, protected time, space, equipment access, and sustainability

Up to 1500 chars

ChemicalQDevice will provide protected PI effort, sponsor oversight, controlled repositories, source and model versioning, assurance-case maintenance, records stewardship, and release governance. Clinical space, operating-room time, perioperative beds, pharmacy, laboratories, imaging, pathology, simulation facilities, robotic hardware, and emergency resources must be secured through written institutional agreements before activation. No field in this draft should be read as evidence of an existing hospital commitment.

9 Budget, Milestones, and Sustainability

Agency-specific caution. RFA-RM-27-001 instructs applicants to enter only the five-year total direct costs and states that a detailed budget and other budget forms are not required and will not be accepted. Use the planning fields below for funder outreach or other sponsors, but omit any module prohibited by the target solicitation.

Requested direct costs / year

Total direct costs

Cost sharing

Indirect cost treatment

Program income

Currency

High-level use of funds and cost rationale

Up to 1650 chars

Funds will primarily support protected PI effort; regulatory and clinical-operations personnel; biostatistics and data management; EDC, secure storage, and cybersecurity; investigational pharmacy and specimen handling; site startup, training, and simulation drills; monitoring and independent medical oversight; patient screening and follow-up support; and repository curation, archiving, and dissemination. This cost structure is appropriate because the project's central challenge is safe, verifiable translation of a complex early-phase oncology platform, not basic capital acquisition alone [NIH, 2026a; Kawchak, 2026c; Kawchak, 2026b].

Milestones, quantitative success criteria, decision gates, and go/no-go thresholds

Up to 2000 chars

Milestone 1: complete registrations, final Combined IND/IDE and IRB determinations, ClinicalTrials.gov setup, site agreements, protocol/software freeze, and successful Phase 0 and emergency drills. Milestone 2: open the first 3+3 cohort only after all verification-before-generation gates, force/timing controls, operator credentialing, and audit-chain tests pass. Milestone 3: progress through the 160, 220, and 300 mg levels, enrolling up to 18 treated participants only after documented cohort review. Success criteria are an interpretable MTD/RP2D decision; complete 30-day device/procedure serious-event ascertainment; assigned robotic tasks completed without unacceptable unsafe conversion; at least 95% completeness for required primary-window data; reproducible provenance for every patient-facing generation; and no uncontrolled cyber-physical failure. Stop, redesign, or de-escalate for a treatment-related death, essential safety-override failure, repeated uncorrectable gate or device failure, unacceptable toxicity, broken auditability, or accrual below 50% of plan for two consecutive quarters.

Five-year timeline and major deliverables

Up to 1850 chars

Year 1: registrations, FDA/IRB pathway, contracts, ClinicalTrials.gov, site qualification, data validation, assurance-case review, and activation. Years 2–3: sequential 3+3 enrollment of up to 18 treated participants, cohort review, and 30-/90-day safety follow-up. Years 3–4: complete 24-month survival follow-up as participants mature, adjudicate endpoints, lock the database, and decide proceed/redesign/stop. Year 5: publish results, release permitted data, code, figures, and provenance artifacts, maintain regulatory records, and prepare follow-on work only if Phase 1 evidence supports it.

Risk register: scientific, enrollment, regulatory, operational, data, supply, and financial risks

Up to 1750 chars

Scientific risk: unacceptable toxicity or infeasible integration of drug and robotic procedure; mitigation: staged safety review and conservative stopping rules. Enrollment risk: slow identification of eligible KRAS G12-mutated resectable PDAC; mitigation: molecular prescreening, tumor-board referral, and backup referral pathways. Regulatory risk: delay in IND/IDE/IRB clearance; mitigation: early submission sequencing and tracked deficiency responses. Operational risk: credentialing gaps, OR scheduling conflict, or device downtime; mitigation: activation checklist, backup scheduling, spare components, and dry runs. Data risk: incomplete source capture, privacy breach, or unstable software versions; mitigation: EDC controls, audit trails, RBAC, backups, and CAPA. Supply risk: drug, assay, or hardware availability constraints; mitigation: qualification of vendors, inventory thresholds, and contingency contracts. Financial risk: underestimation of startup, monitoring, or secure-data costs; mitigation: reserve margins, phased spending, and milestone-based reforecasting.

Sustainability, dissemination, scale-up, follow-on funding, and post-award transition

Up to 1750 chars

Sustainability will come from converting the award's one-time platform build into reusable regulated assets: validated SOPs, repository-based workflows, analysis code, data dictionaries, safety-governance templates, and a decision-ready Phase 2 package [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026d]. Dissemination will include peer-reviewed manuscripts, ClinicalTrials.gov reporting, controlled-access data release, public protocol-adjacent materials where appropriate, and conference presentations targeted to oncology, surgery, regulatory science, and biomedical AI communities. If Phase 1 milestones are met, scale-up will focus on a comparative randomized study, broader site qualification, and possibly platform adaptation to additional pancreatic or other solid-tumor settings. Follow-on support could include later-stage NIH mechanisms, NCI-aligned cooperative structures, philanthropy, or mission-aligned non-dilutive translational funding. Post-award transition planning will preserve records, repositories, and governance roles so that no critical knowledge is lost between mechanisms.

10 Investigator Profile, Innovation Record, and Current Support

Preparation note. The public-profile defaults below were drawn from the ResearchGate profile supplied by the user. Verify titles, affiliations, contact details, publication metrics, registrations, and current/pending support before every transmission. For the NIH Pioneer baseline, only a single PD/PI is allowed

Name	Kevin Kawchak
Degrees	M.S. (Chemistry)
Current position	Chief Executive Officer
Institution / organization	ChemicalQDevice
Email	kevink@chemicalqdevice.com
ORCID	https://orcid.org/0009-0007-5457-8667
eRA Commons ID	Pending; active PD/PI account required.
ResearchGate	https://www.researchgate.net/profile/Kevin-Kawchak
GitHub	https://github.com/kevinkawchak
Expertise	Artificial intelligence; oncology clinical trials; chemistry

Professional profile and qualifications relevant to the proposed project

Up to 1700 chars

Kevin Kawchak has a recent public portfolio spanning IND drafting, Phase 1 and Phase 2 pancreatic cancer protocol development, repository-based LLM document generation for oncology-trial workflows, and PDAC simulation work including digital-twin, QSP, and large-scale in silico trial studies. This body of work is consistently centered on end-to-end oncology trial design, regulatory documentation, verification focused AI methods, and translation of computational workflows into auditable clinical-trial materials. [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026c; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025e] Mr. Kawchak has also published works related to Physical AI oncology legislation [Kawchak, 2026s; Kawchak, 2026t; Kawchak, 2026u; Kawchak, 2026v; Kawchak, 2026w] that supports this clinical trial; with corresponding narrative reviews [Kawchak, 2026x; Kawchak, 2026f; Kawchak, 2026y]. Guidance for oncology and artificial intelligence was created to assist the Physical AI oncology trial transition: <https://github.com/kevinkawchak/LLMs-Pharmaceutical/tree/main/Guidance>.

Selected evidence of creativity, innovation, cross-disciplinary work, leadership, and execution

Up to 2050 chars

Selected evidence of innovation includes public posting of a Phase 1 IND package, companion Phase 1 and Phase 2 protocol documents, repository-based methods for accelerated generation of IND/IRB/amendment/CSR-style materials, and PDAC simulation studies that combine digital-twin, QSP, and large-scale in silico trial approaches with verification, validation, and uncertainty quantification. The portfolio is notably cross-disciplinary across oncology, robotics-adjacent trial concepts, regulatory writing, and computational modeling. [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026c; Kawchak, 2026d; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2026f]

Selected publications, preprints, software, datasets, patents, protocols, or other products

Up to 1900 chars

Investigational New Drug Application - Daraxonrasib, Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC [Kawchak, 2026a]; Phase 1 Pancreatic Cancer Trial: Efficient LLM Document Guidance [Kawchak, 2026b]; Oncology Trial PI LLM Adoption Guide [Kawchak, 2026e]; On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma [Kawchak, 2026c]; A Phase 2, Daraxonrasib + LLM Guided Robotic PDAC Whipple Procedure, Clinical Trial Protocol [Kawchak, 2026d]; QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook [Kawchak, 2025a]; ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate [Kawchak, 2025b].

Current and pending support, source, project, period, total amount, and person-months

Up to 2100 chars

Based on the materials supplied for this draft, no active federal or foundation award with verified project number, budget, and committed person-months was provided for inclusion. Current activity is best described as internally directed ChemicalQDevice research and development spanning 2024-present, including IND drafting, Phase 1 and Phase 2 protocol development, oncology-trial document-generation methods, and PDAC simulation studies [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026c; Kawchak, 2026d; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025e]. Pending support: this NIH Director's Pioneer Award application, 09/01/2027-08/31/2032, requested direct costs \$3,500,000 total (\$700,000/year), PI effort >6 person-months in years 1-3, 4 person-months in year 4, and 3 person-months in year 5 [NIH, 2026a]. If any additional current or pending sponsored projects exist, replace this text with the exact sponsor, project title, period, total costs, overlap statement, and person-months before submission.

Collaborators and institutional affiliations (include only where permitted by target funder)

Up to 1450 chars

The RFA permits only a single PD/PI in the application profile. Kevin Kawchak, ChemicalQDevice, is the proposed PI. Required functional collaborators include pancreatic surgical oncology, medical oncology, anesthesia, pharmacy, pathology, radiology, biostatistics, data management, independent safety monitoring, robotics/device engineering, cybersecurity, and clinical monitoring, but no person or institution should be described as committed until documented. Collaborator Biosketches and support letters are not included; any required assignment-request exclusions or permitted essay references will use confirmed names only.

11 Assurances, Compliance, and Attachment Checklist

Applicant certifications and readiness checks

- ☐ Information is accurate, current, complete, and not misleading.
- ☐ Applicant and organization meet the target opportunity's eligibility requirements.
- ☐ Required organizational and investigator registrations are active and identifiers match.
- ☐ Conflicts of interest, other support, foreign components, and required disclosures are addressed.
- ☐ Human-subject, IRB / sIRB, consent, privacy, inclusion, and safety requirements are addressed.
- ☐ IND / IDE and other regulatory determinations are documented where applicable.
- ☐ Data management, sharing, repository, cybersecurity, and retention requirements are addressed.
- ☐ Budget values, indirect-cost treatment, and cost-sharing representations are consistent.
- ☐ The applicant has checked suspension, debarment, responsibility, and integrity requirements.
- ☐ The final application was reviewed against the live opportunity before the deadline.

Baseline RFA-RM-27-001 attachment controls

- ☐ Essay.pdf / Research Strategy is no more than five pages.
- ☐ Science-area codes appear both in the routing field and at the beginning of the essay.
- ☐ Facilities and Other Resources is no more than one page.
- ☐ Only the PD/PI biographical sketch is included.
- ☐ Current and Pending Support includes all sources and person-month effort.
- ☐ The required 2026-format DMS Plan is included when scientific data will be generated.
- ☐ At least 1 human study record or applicable delayed-onset record is included when required.
- ☐ Exactly three reference letters are arranged for separate, timely submission.
- ☐ No prohibited Specific Aims, separate bibliography, equipment attachment, collaborator biosketches, letters of support/collaboration, or detailed budget forms are included.

Organization-specific assurances, representations, certifications, and exceptions

Up to 1550 chars

Before submission, verify SAM/UEI/CAGE, Grants.gov, organizational and PI eRA Commons registrations, PI ORCID, distinct PI/SO accounts where applicable, and consistency across forms. The application must use the RFA's required five-page Essay.pdf headings and page limit, omit a Specific Aims attachment and detailed budget forms, include Current and Pending Support, and arrange exactly three reference letters through the separate process [NIH, 2026a; NIH, 2026b].

Because the project generates scientific data and is a clinical trial, the final package also requires the structured Data Management and Sharing Plan, Human Subjects and Clinical Trials study record, inclusion information, recruitment and retention plan, protection of human subjects, and monitoring plan. No enrollment may begin without documented FDA/IDE status as applicable, IRB approval, ClinicalTrials.gov compliance, informed consent, site authorization, and validated human-in-command safety controls.

Final attachment inventory and file names

Up to 1550 chars

Components will be provided at a later date: Essay.pdf; Facilities and Other Resources; PD/PI Biosketch; Current and Pending Support; structured Data Management and Sharing Plan; PHS Human Subjects and Clinical Trials Study Record and required study attachments; Foreign Justification only if applicable. Three reference letters are submitted separately. Do not include Specific Aims, a bibliography attachment, an equipment attachment, collaborator Biosketches/support letters, or detailed budget forms unless the live RFA changes.

Unresolved issues, responsible owner, and deadline for resolution

Up to 1450 chars

Open items before submission: activate and reconcile registrations; document separate PI and SO eRA accounts if the same person holds both roles; secure the clinical site and all functional agreements; confirm FDA IND/IDE and IRB pathways; validate the final model, repository commit, sensor interface, safety gates, and emergency controls; finalize the DMS and clinical-trial study records; verify up to 18-participant accrual feasibility; obtain three independent reference letters; and run a final RFA, page-limit, PDF, accessibility, and form-validation review. Owners are the PI/Sponsor-Investigator, Signing Official, regulatory lead, site PI/surgeon, statistician, data steward, device/AI safety lead, and grants administrator, with resolution required before the September 9, 2026 deadline.

Certification

I certify that I have reviewed this application package and that the information presented is ready for the intended funder-facing use, subject to institutional authorization and the target funder's official requirements.

Kevin Kawchak

Signature / typed certification

July 12, 2026

Date

Template Notes and Source Record

This appendix is guidance for the drafter and should normally be removed from a final funder-facing application by setting `\TemplateAppendixfalse` in `main.tex`.

- The baseline opportunity is the NIH Director's Pioneer Award, RFA-RM-27-001, a DP1 opportunity that permits clinical trials and emphasizes unusually innovative, broadly impactful research and a substantial change in research direction [NIH, 2026a].
- The NIH general page-limit table lists a five-page Research Strategy for DP1 applications, while the opportunity limits Facilities and Other Resources to one page [NIH, 2026b; NIH, 2026a].
- The opportunity's required essay headings are reproduced as separate bounded pages in Section 3; it also requires specific research-effort commitments and, for clinical trials, use of the PHS Human Subjects and Clinical Trials information structure [NIH, 2026a].
- The data-sharing module follows the 2026 NIH pilot DMS Plan's structured elements, including yes/no questions, a limitation rationale, and data-type / repository entries [NIH, 2026c].
- Public profile defaults for Kevin Kawchak were taken from the user-supplied ResearchGate profile, which lists an M.S. in Chemistry, Chief Executive Officer at ChemicalQDevice, and a GitHub link [Kawchak, 2026z]. [Kawchak, 2026a], [Kawchak, 2025a], [Kawchak, 2025b]

Important limitations

The character caps in this template are drafting controls, not official agency validation rules. PDF form text wraps and auto-sizes within a fixed-height field. The widget remains on its assigned page and does not flow to the next page; a PDF viewer may clip, stop accepting input at the maximum length, or provide internal scrolling depending on the viewer. Always generate a flattened, visually reviewed final PDF when the recipient requires a non-editable attachment.

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Citation: Kawchak K. Clinical Trial Funding Application v2.0, RFA-RM-27-001, Kawchak K. Zenodo. 2026; 10.5281/zenodo.21317266.